

FYP1 PRESENTATION

University FAQ Chatbot with Open source LLMs

Student Name: HAROLD GOH
Supervisor: Dr. Prabha Kumaresan
Moderator: Dr. Syed Qamrun Nisa

CONTENTS

- | | | | |
|----|--|----|------------------------------------|
| 01 | <u>Introduction</u> | 07 | <u>Background Study</u> |
| 02 | <u>Project Overview</u> | 08 | <u>Methodology</u> |
| 03 | <u>Problem Statements & Objectives</u> | 09 | <u>System Design</u> |
| 04 | <u>Motivations</u> | 10 | <u>Implementation Plan</u> |
| 05 | <u>Justification</u> | 11 | <u>Risks & Mitigation Plan</u> |
| 06 | <u>Project Scope</u> | 12 | <u>Prototype Demo</u> |

INTRODUCTION



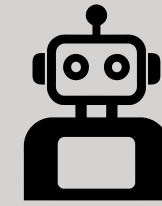
Current Challenge

Traditional Q&A systems lack personalized context and struggle with academic terminology.



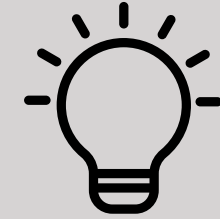
Student Pain Point

Students face information overload without proper guidance or content filtering.



Technology Gap

Existing chatbots provide generic responses without domain-specific knowledge.



New Opportunity

LLMs with RAG offer intelligent, contextualized student support.

PROJECT OVERVIEW

Chatbot Development

- **Automation of Inquiries:**

The chatbot is designed to automate responses to common student inquiries.

- **Improving User Experience:**

Enhancing user experience with quick access to admission-related information.

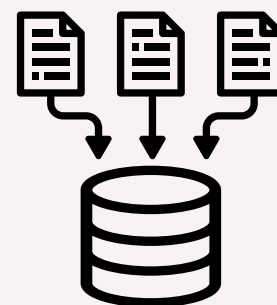
Utilization of Open source LLMs

- **Leveraging Open-source Technology:**

Employing open-source LLMs like Mistral to generate precise responses.

- **Addressing Common Challenges:**

Tackling challenges like hallucinations in AI responses using RAG.



Designing a User Friendly Interface

- **User-Centric Design:**

Focusing on a design that prioritizes the needs and frustrations of users.

- **Web Interface Development:**

Creating a responsive web interface using React for seamless engagement.

- **Testing and Feedback:**

Implementing user testing to gather insights on interface usability.

PROBLEM STATEMENTS & OBJECTIVES

PROBLEMS IDENTIFIED:

- Problem 1: Inefficiency in handling repetitive inquiries.
- Problem 2: Limited interactivity of static information sources.
- Problem 3: Lack of contextual accuracy response in existing systems.



OBJECTIVES:

- Objective 1: To develop a interactive University FAQ chatbot
- Objective 2: To create a user friendly web interface that enables interaction between users and chatbot
- Objective 3: To implement RAG architecture that ensures responses are both accurate and natural.



MOTIVATIONS



1

Addressing Student Frustration

Navigating massive PDF handbooks and static FAQ websites often leaves students feeling lost and frustrated.

2

Increase Efficiency for Staff

Implementing AI helps reduce staff workloads, especially during peak admission seasons.

3

Advancements in AI Technology

Unlike static pages, AI provides a natural, conversational way for users to clear their doubts

JUSTIFICATIONS



Inefficiency of Current Systems:

Traditional methods like static FAQs and manual emails are failing to meet the "instant and accurate" expectations of the digital era.

Administrative Burden:

There is a critical need to reduce staff workloads during peak seasons when repetitive inquiries regarding enrollment and fees overwhelm university offices.

Reducing Hallucination:

Implementing RAG is justified as a necessary solution to overcome the common problem of AI "making things up" by grounding it in trusted university documents.



PROJECT SCOPE

Integration of Open-Source LLMs

- **Model Selection:**

The project utilizes lightweight and capable open-source models from Hugging Face

- **Local Execution:**

Models are chosen to ensure efficient performance and execution on local hardware.

Development of the RAG Pipeline

- **Framework:**

The pipeline is built using the **LangChain** framework to orchestrate the flow of data

- **Contextual Retrieval:**

The system manages the retrieval of relevant information from provided university sources.



Web Interface Development

- **User Accessibility:**

A dedicated web interface is developed to allow students to interact with the chatbot easily

BACKGROUND STUDY

Category	Key Findings & Development	References
Evolution of Chatbots	Early chatbots relied on NLP pipelines, intent classification, and predefined responses, which struggled with unstructured queries.	(Singhuang & Natho, 2025)
LLM Capabilities	Modern models like GPT-3.5 and Llama-3 show strong potential for multilingual support and can extract information from unstructured academic documents.	(Atiq, Ubakanma, & Iqbal, 2025)
The Hallucination Problem	LLMs can produce hallucinations due to data bias or limited context windows.	(Reddy, Kumar, & Prakash, 2024)
RAG Architecture	Retrieval Augmented Generation (RAG) grounds AI in trusted documents (e.g., handbooks), replacing parametric memory with factual retrieval to stop errors.	(Joshi et al., 2024)
Evaluation Frameworks	The RAGAS and ARES frameworks provide automated, scalable methods to evaluate faithfulness, relevance, and context recall in AI systems.	(Es et al., 2024)

METHODOLOGY

```
# 1. extract text from pdf
def extract_text_from_pdf(pdf_path):
    reader = PdfReader(pdf_path)
    text = ""
    for page in reader.pages:
        text += page.extract_text() + "\n"
    return text

# 2 chunk the text
def chunk_text(text, chunk_size=1000, overlap=200):
    chunks = []
    for i in range(0, len(text), chunk_size - overlap):
        chunks.append(text[i : i + chunk_size])
    return chunks

#2: store to chromadb
# Initialize Client
client = chromadb.PersistentClient(path="./my_vector_db")
collection = client.get_or_create_collection(name="pdf_collection")

# Load and Process
pdf_text = extract_text_from_pdf("C:\\Users\\Harold goh\\Download\\scholarship.pdf")
text_chunks = chunk_text(pdf_text)

# Create IDs for each chunk
ids = [f"doc1_chunk{i}" for i in range(len(text_chunks))]
metadatas = [{"source": "scholarship.pdf", "chunk_index": i} for i in range(len(text_chunks))]
```

Three-Phase RAG Architecture:

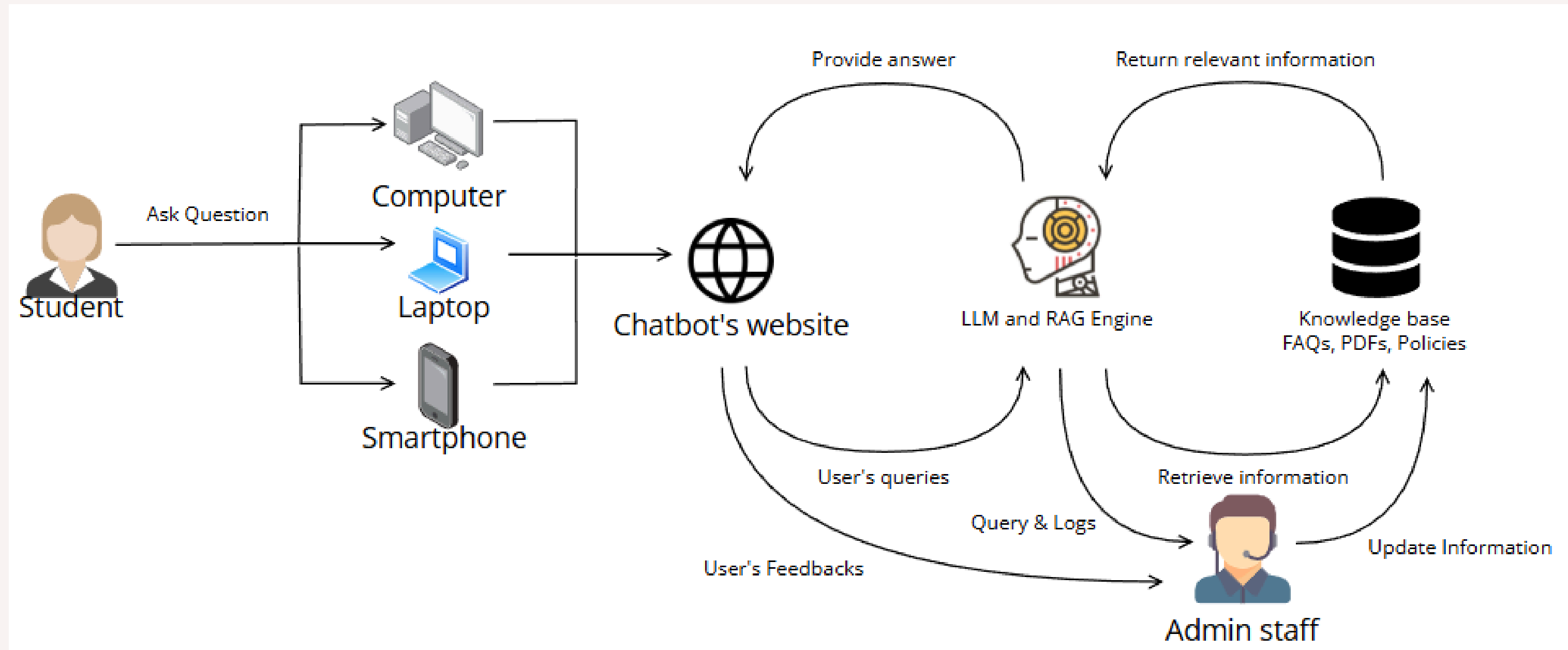
DATA PREPARATION PHASE

INTELLIGENT RETRIEVAL PHASE

GENERATIVE RESPONSE PHASE

- The unstructured data (handbooks, PDFs) is processed and chunked into manageable segments.
- The chunked data is encoded into vector embeddings using the Sentence-BERT model and stored in a vector database
- The LangChain framework acts as middleware to manage the interaction between user input, the vector store, and the LLMs (Mistral 7B, Llama 3.1, Phi3.8b)

SYSTEM DESIGN



The System Core: At the center of the architecture is the LLM and RAG Engine, which handles all necessary processing to generate responses. The engine retrieves factual information from the knowledge base that consists of university FAQs, PDF handbooks, and official policies.

IMPLEMENTATION PLAN

This phase focus on three main areas:

- Front-end
- Back-end
- RAG Pipeline

DEVELOPMENT PHASE

To ensure the system meets user expectations and technical requirements:

TESTING PHASE

INTEGRATION PHASE

This phase ensures all components work as a unified system:

DEPLOYMENT PHASE

Moving the system from development to a production environment

RISKS & MITIGATION PLAN

Risks	Mitigation Plan
Computational Constraints	Utilizing lightweight models to ensure efficient performance on local machines.
Hallucinations and Inaccurate Data	Designing a system prompt to enforce constraints and using similarity validation to ensure answers stay grounded in the retrieved facts.
Outdated Information	Involve admin staff intervention to manually re-index the knowledge base by uploading new PDFs or deleting outdated ones.

Current Device Spec: GPU 4GB, RAM 8GB, CPU AMD RYZEN 5

CONCLUSION

Project Summary

- **Technical Roadmap:**

This project has successfully established a clear design and implementation plan for the proposed chatbot.

- **Solution Strategy:**

The methodology focuses on grounding AI with a trusted knowledge base

Key Contributions

- **System Design:**

Finalized the architecture, including Rich Picture, Use Case, and Class diagrams to guide the FYP2 development phase.

- **Tech Stack Validation:**

Validated the usability of chosen models (Mistral 7B, Phi, and Llama 3.1) on local hardware

Future Work (FYP 2)

- **Development Focus:**

Actual coding of the front-end and back-end as outlined in the implementation plan.

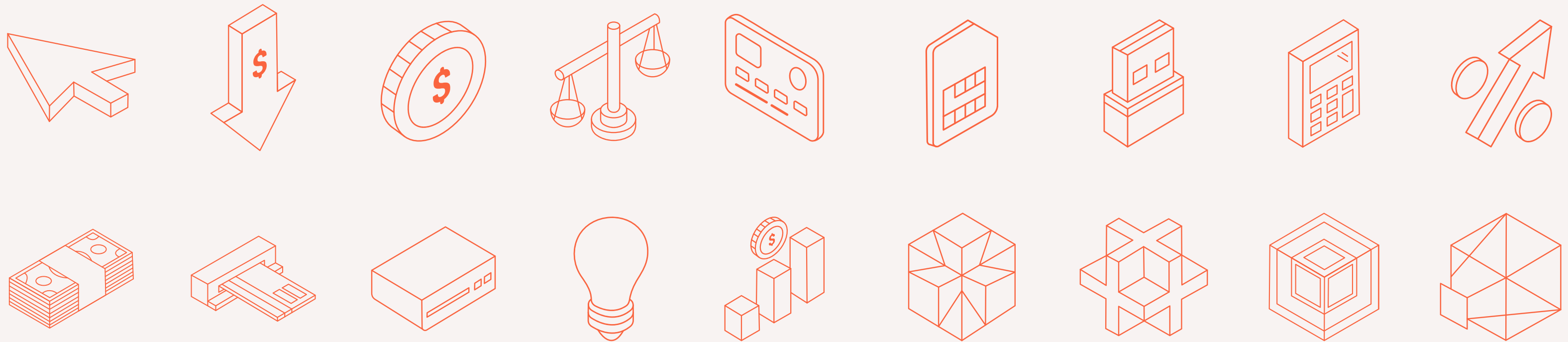
- **Testing & Evaluation:**

Conduct testing using the RAGAS framework to evaluate the accuracy, and compare with the base models.

REFERENCES

- Singhuang, N., & Natho, P. (2025). An Enhanced Chatbot for University Admission and Educational Guidance. IEEE.
- Atiq, A., Ubakanma, G., & Iqbal, M. (2025). Evaluating efficacy of LLMs and Embedding Models in Resume Parsing: A Comparative Analysis Using Open Source LLMs. IEEE. Retrieved from <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=11136269&isnumber=11136249>
- Reddy, G. P., Kumar, Y. V., & Prakash, K. P. (2024). Hallucinations in Large Language Models. Retrieved from <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10542617&isnumber=10542531>
- Joshi, D., Tekade, S., Zanzane, S., Tiwadi, S., Tripathi, A., & Sakharwade, D. (2024). Generative AI-Driven Chatbot for Personalized University Information and Assistance. Retrieved from <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10882421&isnumber=10882170>
- Es, S., James, J., Espinosa-Anke, L., & Schockaert, S. (2024). RAGAS: Automated Evaluation of Retrieval Augmented Generation. Association for Computational Linguistics. Retrieved from <https://arxiv.org/abs/2309.15217>

PROTOTYPE DEMO



THANK YOU